

Intelligent Traffic Monitoring and Congestion Prediction System Using Computer Vision

1. Vehicle Detection and Traffic Monitoring

The Vehicle Detection and Traffic Monitoring module is the core of the proposed system. It captures live video from roadside cameras and processes the footage with Computer Vision techniques to detect, classify, and track vehicles in real time. Artificial Intelligence examines the video to identify various types of vehicles, including cars, buses, trucks, motorcycles, and bicycles. Deep learning models effectively detect vehicles in different lighting and weather conditions, while object tracking algorithms follow vehicle movement across consecutive video frames. The system counts vehicles, measures traffic flow, and checks lane occupancy to provide real-time traffic data. IoT-enabled devices like Raspberry Pi and cameras enable continuous monitoring and send traffic information to a centralized dashboard for remote access and smart traffic management.

Algorithms Used

- YOLOv8 Nano (YOLOv8n)
- OpenCV
- Deep SORT
- Convolutional Neural Network (CNN)

Conclusion

The Vehicle Detection and Traffic Monitoring module delivers precise, real-time traffic data by leveraging advanced Computer Vision and Artificial Intelligence to detect and track vehicles. By integrating IoT devices, the system ensures continuous monitoring and seamless data transmission, establishing a robust foundation for congestion prediction and intelligent traffic management.

2. Traffic Density Estimation

The Traffic Density Estimation module analyzes detected vehicles to estimate current traffic density across various road segments. Artificial Intelligence assesses parameters such as vehicle count, lane occupancy, traffic flow, and vehicle spacing to categorize traffic conditions into low, medium, or high density. Computer Vision continuously monitors road conditions, updating traffic information in real time. Machine Learning models process both historical and live traffic data to enhance density estimation accuracy. The estimated traffic density is transmitted via IoT-enabled devices to a cloud platform, empowering traffic authorities to monitor congestion remotely and respond efficiently.

Algorithms Used

- Convolutional Neural Network (CNN)
- K-Means Clustering
- Density Estimation Algorithms
- OpenCV

Conclusion

The Traffic Density Estimation module accurately measures road congestion by analyzing vehicle movement and distribution. AI-based estimation improves traffic monitoring and supports effective traffic management by providing reliable real-time density information.

3. Traffic Congestion Prediction

The Traffic Congestion Prediction module forecasts future traffic conditions by analyzing both historical records and real-time traffic density. Artificial Intelligence detects traffic patterns and predicts congestion before it happens, enabling authorities to implement preventive measures. Long Short-Term Memory (LSTM) networks capture temporal traffic behaviors, while Random Forest models assess congestion severity based on traffic flow and vehicle density. Time-Series

Forecasting techniques utilize past observations to estimate future traffic conditions. Predicted congestion data is presented via an IoT-enabled dashboard, supporting intelligent transportation planning and helping to reduce travel delays.

Algorithms Used

- Long Short-Term Memory (LSTM)
- Random Forest
- Time-Series Forecasting
- Gradient Boosting

Conclusion

The Traffic Congestion Prediction module enables early prediction of traffic jams by analyzing historical and real-time traffic data. Artificial Intelligence improves prediction accuracy and supports proactive traffic management, reducing congestion and improving transportation efficiency.

4. Smart Traffic Signal Control

Abstract

The Smart Traffic Signal Control module automatically adjusts traffic signal timing based on real-time traffic conditions. Artificial Intelligence continuously analyzes vehicle counts, queue lengths, waiting times, and traffic density to determine the optimal signal duration for each lane. Reinforcement Learning enables the system to improve signal timing decisions by learning from changing traffic patterns. Optimization algorithms reduce unnecessary waiting time, improve vehicle movement, and minimize fuel consumption. IoT-enabled controllers communicate with traffic signals and receive live traffic information from roadside cameras, enabling adaptive traffic management in smart city environments.

Algorithms Used

- Reinforcement Learning
- Q-Learning
- Genetic Algorithm
- Optimization Techniques
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Conclusion

The Smart Traffic Signal Control module dynamically adjusts traffic lights based on real-time traffic conditions. AI-based optimization reduces congestion, minimizes vehicle waiting time, and improves overall road efficiency through intelligent signal control.

5. IoT-Based Traffic Monitoring Dashboard

Abstract

The IoT-Based Traffic Monitoring Dashboard provides real-time visualization of traffic conditions collected from roadside cameras and IoT devices. The dashboard displays vehicle count, traffic density, congestion level, average vehicle speed, and traffic signal status through interactive charts and graphical reports. Artificial Intelligence continuously updates traffic information by analyzing incoming video data and sensor readings. IoT communication protocols enable secure and efficient transmission of traffic information from Raspberry Pi devices to cloud platforms, allowing traffic authorities to monitor multiple locations from a centralized interface remotely. Historical traffic data is stored for future analysis and congestion prediction.

Algorithms Used

- MQTT Protocol
- IoT Communication
- Data Visualization
- Cloud Computing

Conclusion

The IoT-Based Traffic Monitoring Dashboard enhances intelligent transportation by providing real-time traffic information through cloud-connected IoT devices. The integration of AI and IoT improves traffic monitoring, supports remote decision-making, and enables efficient urban traffic management.